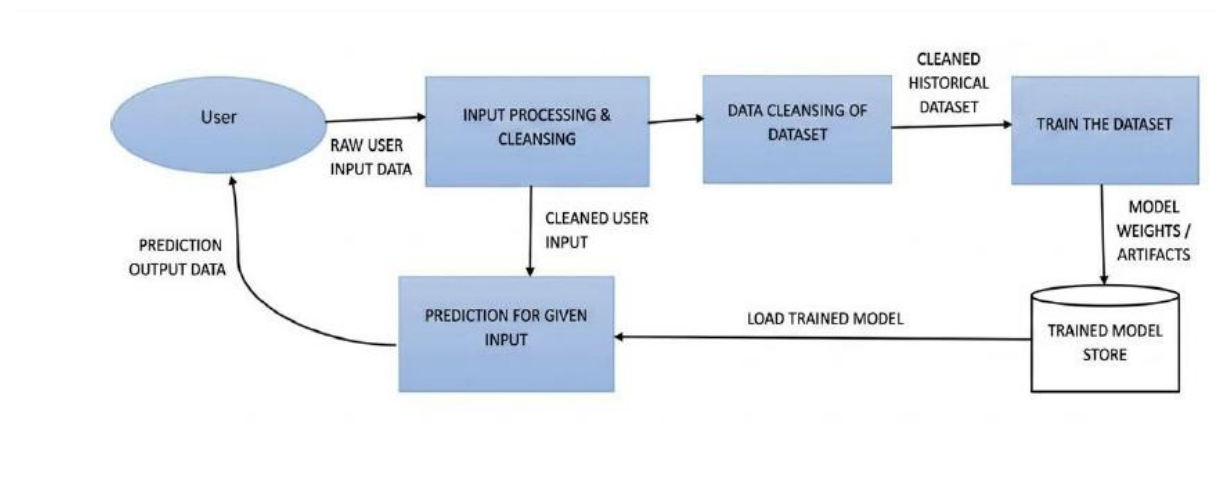


RIISING WATERS:FLOOD PREDICTION SYSTEM USING ML

PROJECT DESCRIPTION:

Floods are among the most devastating natural disasters, causing significant damage to human life, agriculture, infrastructure, and the economy. Predicting flood events in advance enables governments and disaster management authorities to implement preventive measures, reducing the impact on affected communities. This project presents an **Flood Prediction System using ml** that leverages the **XGBoost (Extreme Gradient Boosting)** machine learning algorithm to predict the likelihood of flooding based on environmental and meteorological factors. The system is trained on historical flood data containing various parameters such as rainfall, river levels, drainage conditions, climate factors, and other relevant indicators. After training, the model analyzes input parameters and classifies whether flood conditions are likely to occur. The trained model is saved using Python's Pickle library and can be integrated into a web application or decision support system for real-time flood prediction.

TECHNICAL ARCHITECTURE:



System Architecture Description

The proposed architecture follows a modular pipeline in which historical environmental data is collected and preprocessed before being used to train an XGBoost machine learning model. The trained model is evaluated using multiple performance metrics to ensure its reliability and then stored for future use. During operation, users provide current environmental parameters, which are processed by the saved model to predict the likelihood of flooding. The prediction results are displayed through the user interface,

enabling disaster management authorities to make timely and informed decisions for flood preparedness and response.

PRE-REQUISTES:

S.No	Software/Library	Version (Recommended)	Purpose	Download / Official Link
1	Python	3.10 or later	Programming language used for development	Python Official Website
2	Visual Studio Code	Latest	Source code editor	Visual Studio Code
3	Pandas	Latest	Data loading and preprocessing	Pandas Documentation
4	XGBoost	Latest	Machine learning algorithm for flood prediction	XGBoost Documentation
5	Scikit-learn	Latest	Model training, validation, and evaluation	Scikit-learn Official Website
6	Pickle	Built into Python	Save and load the trained model	Python Pickle Documentation
7	CSV Dataset	-	Historical flood data for training	Project dataset (flood.csv)
8	Windows 10/11	64-bit	Operating system	Microsoft Windows